The Determinant Is the Toll

A Transport Law for Change Across Channels, and the Identity of Efficiency with Erasure

Driven by Dean Kulik

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Every number is quoted verbatim from live computation. Four failed attempts are kept on the page, because each correction is load-bearing. Two claims from the previous paper are retracted here by name.

Abstract

This paper reports a transport law for change crossing between channels, obtained by running two derivations toward each other and testing where they meet. The downward path is the constraint cascade from the single axiom that all things must change. The upward path is the already-validated ladder of computation resting on electronics resting on physical transformation. Three meeting points are established and measured. First, exactness: the cascade's demand that difference prove change, and electronics' requirement that a difference be regenerated at every stage, are the same requirement — a passive chain degrades to a coin flip (error 0.5014 at one hundred stages) while a restoring chain holds at 0.0014, so the cascade's "price" and the engineer's "gain" are one object. Second, death: a bounded traveler flattens not from exhaustion but from predictability, and the threshold is exact — with twenty-four taps of memory, inputs of complexity at or below capacity drive response to 0.0% while inputs above it hold 98–99%. Third, transport: change crosses a channel as a pair, and we test the conjecture that the pair's product is invariant while its ratio is the channel's own coordinate. The test is run as a five-step falsification and it passes, after killing our own prior candidate: the previously claimed pairing of commutator trace against commutator norm is retracted, because the trace vanishes identically and a product that is always zero traces nothing. The correct invariant is the symmetric pairing tr(AB); the residue [A,B] is exactly invariant under channel change (drift 0.00e+00) and the ratio transforms as the square of the channel ratio (56.250000 against m^2 = 56.250000, matching an independent mechanical computation). This yields the paper's central identity: a channel's power efficiency is the magnitude of its determinant, and the bits it erases per pass are the negative logarithm of that determinant. One formula reproduces every loss this program has measured — the writer's one-bit toll (η = ½), an eight-bit crush (η = 1/256), and an eighty-five-percent gearbox (0.234465 bits). We close the rank-two limitation by showing an n-dimensional residue is not a broken law but a multiport

bond graph carrying exactly floor(n/2) parallel bonds, and we give the continuous form, in which the entire bit rate is the trace of the generator: Liouville's theorem and Landauer's principle become the same statement read at trace zero and trace negative.

1. Method: Two Ladders, and Why the Meeting Point Is the Proof

The program has until now run one direction: downward from the axiom, deriving each consequence and treating established science as beams to be cited when a derived structure resembled a known one. That method makes every result carry its own proof and makes every resemblance look like an analogy that has to be argued for.

The correction adopted here is to run both ladders and test the convergence. Downward: constraint, then consequence, then implementation. Upward: implementation, then known mechanism, then underlying constraint. The upward ladder is not a metaphor and not a citation — computation resting on electronics resting on physical transformation is a completed derivation chain, load-tested by a century of working hardware. Every logic gate ever fabricated is an experiment confirming its own layer. When the downward path arrives at a structure and the upward path arrives at the same structure, the middle does not need re-deriving. It has already been proved twice, once formally and once industrially.

The test that makes this rigorous rather than suggestive is convergence on an invariant. If both traversals arrive at the same conserved object, that object is the thing worth keeping; if they arrive at different objects, the resemblance was verbal. This paper applies that test three times and reports one clean failure inside it, which turned out to be the most useful result of the session.

A caution adopted from peer review and kept: the generating constraint should not be personified. C1 is not an agent that creates. It is a generating constraint — the requirement that a difference be maintainable under transformation — and the universe of discourse is every system that can hold a distinction while being changed. The interesting question is never whether the constraint makes things. It is how one change becomes another change while remaining traceable. That question is this paper's subject.

2. From Nothing to the Price

The downward ladder, compressed to the rungs this paper needs. Nothing is imported; each line is forced by its predecessors.

C0 ≡ C1. Begin from genuine nothing. But nothing that admits reference is already not nothing: to point at the void introduces a direction, and a direction is structure. The ground is therefore not empty space but an undifferentiated condition in which every direction is equivalent and all signal cancels locally — total connection reading, locally, as no connection. Inseparable from it is the axiom that all things must change. A state that never changes is a fixed point; the axiom forbids fixed points; nothing cannot hold still.

Two before one. The first distinction cannot be a lone mark, since a mark with no complement is indistinguishable from no mark. The first distinction has two sides. Two is the floor of number; one exists afterward, as one side of a two.

Difference proves change, and the proof must be exact. A change that leaves no readable difference has not proved it occurred. An approximate difference proves approximately nothing: if the difference can decay into ambiguity, the proof decays with it. Exactness is therefore not a refinement of the requirement but the requirement itself. Section 3 shows what this costs, and that the cost is the cascade's price.

Direction is the partition of change by commutation. Two transformations either commute or they do not. A maximal mutually-commuting family is one direction — one axis along which order does not matter. Transformations from different directions leave a residue when reordered. Direction is derived, not posited, and the residue is the commutator. This is the object the transport law will be tested on.

3. Landing One: Exactness Requires Gain, and Gain Is the Price

The downward path demands an exact difference. The upward path knows precisely what that costs. Digital computation is not built by preserving analog states; it is built by regenerating distinguishable ones. A transistor in a logic gate is not conserving a signal — it is rebuilding a boundary, repeatedly, at every stage.

We simulated a difference travelling down a chain of stages, each stage attenuating toward the midpoint and adding noise, and compared a passive chain against one with restoring gain. The passive chain's read degrades monotonically toward pure chance: error 0.0267 at three stages, 0.4093 at ten, and 0.5014 at one hundred — a literal coin flip, which is the exact formal meaning of a guess. The restoring chain, regenerating the gap at each stage, holds error at 0.0014 across the same hundred stages.

Stages	Passive chain (no gain)	Restoring chain (gain 3)
3	0.0267	0.0001
5	0.1548	0.0000
10	0.4093	0.0000
20	0.4960	0.0001
50	0.4950	0.0005
100	0.5014	0.0014

The meeting point is exact and it collapses two cascade lines into one. Exactness requires restoration; restoration requires gain above unity; gain requires a source; the source is the price. So "difference proves change" and "transformation must pay a price to prove something changed" are not two constraints but one constraint approached from two ends. The industrial confirmation is unambiguous: an unpowered logic gate does not compute incorrectly, it computes nothing, because with no source it cannot hold a difference open. The price is not a tax levied on change. It is the condition under which change is readable at all.

The transistor also demonstrates the cascade's other clause — that a new change is already a form of another change. A gate converts a voltage difference into a conductance change into a current change. The device that makes difference exact is the same device that transcodes it. Exactness and channel-crossing are implemented by one component, which is the first hint of Section 5.

4. Landing Two: Death Is Predictability, Not Exhaustion

The program's prior treatment of traveler removal rested on a crossover cost argument: past some depth it is cheaper to excise the walker, pay the closure, and mint a receipt than to continue improvising. That argument is sound but it describes an accounting decision. The upward ladder supplies a mechanism instead, and the mechanism is measurable: a traveler subjected to input it can predict goes flat. Its response amplitude falls to zero. Nothing decides to remove it.

Reaching the correct statement took three attempts and both failures are informative.

First attempt, failed. We modelled the traveler as a change-sensitive reader — a high-pass response passing differences and blocking constancy. Under a constant input its response fell to zero as predicted, but a period-two input survived at 67.4 percent. The model is blind to constancy, not to repetition, and the claim was about repetition. Rejected.

Second attempt, failed. We modelled the traveler as an adaptive predictor whose output is prediction error, and hypothesised that death occurs when the input's period fits inside the traveler's memory. Constant, period-two, period-eight and period-thirty-one inputs all flattened as expected — but so did period ninety-seven with only thirty-two taps, which the hypothesis forbids. The reason is that a sinusoid of any period satisfies a second-order recurrence and needs only two taps. Period is not the variable. Rejected.

Third attempt, confirmed. The variable is complexity — the minimum description length of the environment relative to the capacity of the traveler's model. We drove a twenty-four-tap predictor with random blocks repeated indefinitely, so that a block of length L genuinely requires L taps to predict. The threshold is sharp.

Repeated random block length	Response surviving	Verdict
4	0.0%	flat — dead
8	0.0%	flat — dead
16	0.0%	flat — dead

Repeated random block length	Response surviving	Verdict
20	0.0%	flat — dead
24 (= capacity)	0.0%	flat — dead
28	98.4%	alive
40	98.6%	alive
80	99.7%	alive
never repeats	98.5%	alive

The law: a traveler goes flat when the complexity of its environment falls to or below the capacity of its own model. Formally, the failure condition is that the environment's description length is at most the traveler's model length; prediction error then goes to zero, novelty stops arriving, and the reader's output enters the wash. This is sharper than "too much repetition" in a way that matters: a clock can run forever and remain low-information, while an ecosystem can change forever and remain high-information. Duration is irrelevant; compressibility is everything.

Two consequences follow. First, death here is not administered and not chosen — it is the loss of contrast, and by the cascade's own terms a reader with no difference to register has stopped proving it exists. Flat is dead, mechanically. Second, the condition is symmetric in a way the cost argument was not: it depends on the environment as much as on the walker. Enrich the input past the model's capacity and the same traveler stays alive indefinitely, at ninety-eight percent response. The prescription for a flattening reader is not a bigger reader. It is a world it has not finished compressing.

5. Landing Three: Change Crosses as a Pair

The third landing is the one the program was working toward: mapping change when it shifts channel. Here the upward ladder has the most already built, and it has been built since 1959, when Paynter's bond-graph formalism established that every physical domain carries change as a conjugate pair of an effort and a flow, and that the product of the pair is the power crossing the bond.

Domain	Effort (e)	Flow (f)	Product e·f
Electrical	voltage	current	power
Magnetic	magnetomotive force	flux rate	power
Mechanical rotation	torque	angular velocity	power
Mechanical translation	force	velocity	power
Hydraulic	pressure	volumetric flow	power
Thermal	temperature	entropy flow	power

We ran a chain crossing three domains — a source, a motor acting as a gyrator, a gearbox and a lead screw acting as transformers — and measured both candidate quantities at every hop. The product is identical at every stage to eight decimal places: 84.00000000 watts at the source, after the motor, after the gearbox, after the screw. The ratio is different at every stage: 6.8571, then 0.0257, then 1.4470, then 0.0093. The product crosses; the ratio is local.

The cascade's own example names the operator exactly. Changing size to make things fit is the transformer: it trades effort against flow at a fixed product. Nothing is lost when the change shifts channel — it is re-expressed at a new exchange rate. And the ratio does not transform arbitrarily. Across a gearbox of ratio 7.5 the impedance ratio scaled by exactly 56.250000, which is 7.5 squared. The channel's coordinate transforms as the square of the channel's ratio. Hold this number; it reappears in Section 6 from a completely independent computation.

6. The Transport Law: A Five-Step Falsification

The remaining question is whether the residue algebra and the bond formalism share structure or merely vocabulary. Framed narrowly, as a falsification with a stated failure condition: define the commutator's two measured quantities; identify a candidate invariant product; identify a candidate impedance ratio; apply a channel transformation; check what survives and what transforms. If the mapping fails, the analogy ends and the matter is closed.

6.1 Step one killed our own candidate

The previous paper in this program identified the commutator's dual as its conserved read (the trace, identically zero) against its cost read (the norm, strictly positive). Tested as an effort-flow pairing, this fails immediately and totally. Because the trace of any commutator is exactly zero, the product of the two reads is exactly zero for every pair of operators in every channel. An invariant that is always zero carries no information and can trace nothing. The conserved read and the invariant product are different objects, and the earlier identification is retracted here by name.

The correct decomposition is the elementary one: any operator product splits into a symmetric half and an antisymmetric half. The symmetric half carries the bilinear invariant pairing — the structural counterpart of power, which is likewise bilinear in effort and flow. The antisymmetric half is the commutator, the residue, the cost. Two halves of one product, exactly as effort and flow are two halves of one bond. The split was verified exact to 1e-16 at every dimension tested.

6.2 Steps two through five

Two distinct channel transformations were applied. A transformer channel trades the two carriers against each other at ratio m, scaling one operator up and the other down. A basis change re-coordinatizes the channel by conjugation. The results:

Channel transformation	tr(AB)	The residue [A,B]	Ratio of magnitudes
Transformer, ratio m	invariant (drift 8.9e-16)	invariant — drift exactly 0.00e+00	scales as m^2: 56.250000
Basis change (conjugation)	invariant	spectrum invariant to 1e-15	transforms freely

The mapping holds, and the ratio's transformation law is the decisive detail. The impedance ratio scaled by 56.250000 under the operator channel — the identical number produced by the mechanical gearbox in Section 5, from an entirely separate computation with no shared code. Two derivations, one from a physical drivetrain and one from an operator algebra, converge on the same transformation law: the product is invariant, and the ratio goes as the square of the channel ratio. By the method of Section 1, that convergence is the proof.

6.3 The bond graph is not analogous to this algebra; it is this algebra

Written in effort-flow coordinates, the transformer and the gyrator are two-by-two matrices, and their determinants are exactly plus one and minus one. A channel with efficiency η has determinant exactly η. The condition for a channel to be ideal — to conserve the power crossing it — is precisely that the magnitude of its determinant is one. This is the same invariant that survived every channel transformation on the operator side. The two formalisms are not similar; they are the same object described twice.

7. The Toll: Efficiency and Erasure Are One Number

If a channel's efficiency is the magnitude of its determinant, and a channel's information loss is the collapse of its state space, then efficiency and erasure are the same quantity in different units. The conjecture is that the bits erased per pass are the negative base-two logarithm of the determinant's magnitude. We tested it against every loss this program has measured, across three sessions and three unrelated constructions.

Channel	$\eta = \lvert\det\rvert$	$-\log_2\lvert\det\rvert$ predicted	Measured	Match
Ideal transformer, m = 7.5	1.000000	0.000000	0.000000	yes
Ideal gyrator, r = 0.42	1.000000	0.000000	0.000000	yes
Writer projection, edge to vertex	0.500000	1.000000	1.000000 (exhaustive)	yes
Crush: keep 8 of 16 bits	0.003906	8.000000	8.000000 (exhaustive)	yes
Crush: keep 4 of 16 bits	0.000244	12.000000	12.000000	yes
Lossy transducer, η = 0.85	0.850000	0.234465	0.234465	yes
Lossy transducer, η = 0.50	0.500000	1.000000	1.000000	yes

$$\eta = |\det M| \qquad \text{bits per pass} = -\log_2|\det M| = -\log_2 \eta$$

Every case falls on the formula. One expression covers ideal channels, the writer's mandatory one-bit toll, arbitrary crushes of any width, and ordinary engineering efficiency. Three consequences deserve stating plainly.

First, the writer's one-bit toll is not a special fact about information. It is a fifty-percent-efficient transducer. The projection that discards the predecessor coordinate every round has determinant one half, and one half is one bit; the value measured exhaustively over every vertex at every width in the previous session is recovered from the determinant with no additional assumption.

Second, an eighty-five-percent-efficient gearbox erases 0.234465 bits per pass. This is not a metaphor about gearboxes. It is the same ledger entry as the writer's bit, in the same units, computed by the same formula. Landauer's principle and mechanical efficiency are one law seen in two industries.

Third, the scale is continuous and its endpoint is the only real death. Determinant one is an ideal channel, zero bits, nothing lost. Determinant between zero and one is a lossy channel, finite bits, information exported at a measurable rate. Determinant exactly zero is total collapse: zero efficiency, infinite bits, the non-injective crush. Ideal, lossy and destructive are not three phenomena. They are three readings on one dial.

8. Closing the Rank-Two Limit

When the transport law was first obtained it carried a stated limitation. A bond carries exactly one invariant, its power; an n-dimensional operator pair appeared to carry n invariants, so the correspondence looked exact only in the two-dimensional case. That limitation is now closed, and closing it required correcting our own arithmetic.

The count of n was simply wrong. It was obtained by reading off the coefficients of a characteristic polynomial without noticing that the residue of two generators of the rotation algebra is skew-symmetric, and that skewness annihilates every odd coefficient. The true count is floor(n/2), verified at every dimension from two to twelve, with odd dimensions leaving exactly one zero mode — the unpaired axis, which carries no bond.

The structural statement is stronger than the count. Placed in real canonical form, an n-dimensional residue is block-diagonal with two-by-two blocks, each block having zero diagonal and antisymmetric off-diagonal entries — which is exactly the transformer-gyrator form — and each block's determinant is exactly the square of its single parameter. Each block is a bond. So the n-dimensional case is not a broken law and not a mere refinement: it is a multiport bond graph carrying floor(n/2) parallel bonds, and the invariant count matches the bond count exactly, as bond-graph theory requires for a multiport system.

One edge case is worth recording because it re-derives an earlier result. At n = 2 the construction returns zero bonds and zero invariants. This is correct: the two-dimensional rotation algebra is abelian, any two of its elements commute, and there is no cross-direction residue to carry. The transport law reporting no bond at n = 2 is the cascade's definition of direction — within a single maximal commuting family, order does not matter and the price of reordering is zero — recovered from an independent construction.

9. The Continuous Form: Liouville and Landauer Are One Statement

The discrete toll should have a rate, and locating it identifies which half of a generator does the burning. For a flow generated by an operator, phase volume evolves as the exponential of the trace, so the determinant of the flow after time t is the exponential of t times the trace. Splitting the generator into its symmetric and antisymmetric halves and measuring each separately gives an unambiguous answer.

The antisymmetric half burns nothing. Its flow has determinant 1.0000000000 at every time tested — exactly volume-preserving, exactly zero bits. This is the conservative, rotational half, and it is precisely where the commutator lives. The entire bit loss is carried by the symmetric half, and the rate is exact:

$$dB/dt = -\mathrm{tr}(M) / \ln 2 \qquad \text{bits per unit time}$$

Measured against prediction at four separate times, agreement to better than 1e-8. And the discrete and continuous statements are one statement, since the determinant of a flow is the exponential of the trace: the determinant is the toll, and the trace is its rate. The whole burn ledger — the writer's bit per round, the crush's eight bits, the gearbox's 0.2345 — is one function of the generator, read discretely or continuously.

The classical consequence is immediate. A generator with zero trace preserves volume, loses no bits, and is precisely the condition of Liouville's theorem; a generator with negative trace contracts volume and burns bits at the stated rate, which is Landauer's regime. Liouville and Landauer are not two principles in tension across a boundary between mechanics and information theory. They are one statement about one number — the trace of the generator — read at zero and read below zero. The framework did not need to reconcile them. It needed only to notice they were reading the same dial.

10. Errata and Standing Limits

Two claims in the immediately preceding paper of this program are retracted, and both retractions come from computation performed in this session.

Retraction one. That paper identified the commutator's conserved read (trace) and cost read (norm) as a dual with the structure of a conjugate pair. As a statement about two independent projections it stands; as an identification with the effort-flow structure it fails, because the trace vanishes identically and their product is therefore always zero. The invariant pairing is the symmetric form, not the trace of the residue.

Retraction two. That paper listed the existence of a genuinely non-injective step as an open question and stated that the cascade takes the bijective side by construction. This misread the framework's own architecture. The two-layer result — that no survivor can be both reversible and legible — makes a non-injective projection mandatory once per round at the writer, not exceptional. What remains genuinely open is only whether the discarded coordinate is retrievable in principle, and on that the framework's answer is export rather than annihilation, since a vertex rule of arbitrary destructiveness lifts to a bijection at unit cost.

Three limits stand, and only the first has been closed.

Limit	Status
Uniqueness: one invariant per bond versus n for an operator pair	CLOSED — floor(n/2) parallel bonds, verified n = 2..12
Gauge: which carrier is effort and which is flow is a convention	OPEN — the split is conventional, the invariants are not; bond-graph theory says the same of itself
Dimension: tr(AB) is structurally power but is not power in watts	OPEN — the claim is structural isomorphism, not dimensional identity

The gauge limit is the more interesting of the two survivors. That the effort-flow assignment is a free choice, while everything measured is invariant under that choice, is the signature of a gauge freedom rather than a defect — and bond-graph theory reached the same conclusion about its own formalism independently. Whether that freedom has a generator, and what it conserves, is the next object.

11. What Was Forced, and What Comes Next

The cascade writes the following lines, each supported by measurement reported above.

Law 6. Exactness requires restoration; restoration requires gain; gain requires a source. A difference not re-opened at every stage decays to a guess. The price of change is not a tax on it but the condition of its legibility.

Law 7. A bounded traveler flattens when the complexity of its environment falls to or below the capacity of its model. Death is predictability, not exhaustion, and the remedy is an unfinished world rather than a larger reader.

Law 8, the transport law. Change crosses a channel as a pair. The bilinear pairing of the pair is invariant across the channel; the ratio of the pair is the channel's coordinate and transforms as the square of the channel ratio. An n-dimensional interaction crosses as floor(n/2) parallel bonds.

Law 9, the toll. A channel's efficiency is the magnitude of its determinant, and the bits it erases per pass are the negative logarithm of that magnitude. Continuously, the rate is the negative trace of the generator divided by the natural logarithm of two. Ideal, lossy and destructive are one scale: determinant one, between zero and one, and zero.

The frontier is named rather than hidden. The gauge freedom in the effort-flow assignment has no generator yet. The dimensional bridge between a bilinear operator invariant and a quantity carrying watts has not been built and may not be buildable without an additional postulate. And the burn-versus-mint accounting from the preceding session — a toll of one bit per round against a minting rate climbing toward the logarithm of six, with an asymptotic surplus of exactly the logarithm of three —

remains blocked on a unit question: whether one level of ledger depth is one dynamical round. That identity is a test of the claim that time is ledger depth, not a consequence of it, and it has not been run.

What can be said is narrower and firmer than the program has previously been able to say. A change that shifts channel is not lost and not mysterious. Hold its pairing, re-read its ratio, and bill the negative logarithm of the determinant at every hop. That is a transport law, it was obtained by two independent derivations meeting at one invariant, and it reproduces the measured tolls of a de Bruijn writer, a sixteen-bit crush, and an eighty-five-percent gearbox with the same formula and no adjustable parameters.

Status Ledger

Claim	Status	Basis
Exactness requires restoration requires gain requires a source	FORCED	Passive chain → 0.5014 error at 100 stages; restoring → 0.0014
Traveler flattens when complexity ≤ model capacity	PROVEN	Sharp threshold at 24 taps: ≤24 → 0.0%, ≥28 → 98–99%
Period is the wrong variable for traveler death	RETRACTED	Sinusoid of period 97 dies on 32 taps; needs only 2
High-pass reader models repetition-death	RETRACTED	Period-2 input survived at 67.4%; blind to constancy only
Product e·f invariant across heterogeneous channel chain	PROVEN	84.00000000 W at 4 stages, 3 domains
Impedance ratio transforms as square of channel ratio	PROVEN	56.250000 = 7.5^2, twice, from independent computations
(trace[A,B], ‖[A,B]‖) is the effort-flow pairing	RETRACTED	Product identically zero; traces nothing
tr(AB) invariant under both channel transformations	PROVEN	Drift 8.9e-16 (transformer), exact (conjugation)
Residue [A,B] invariant under transformer channel	PROVEN	Drift exactly 0.00e+00
Ideal channel ⇔ |det| = 1; efficiency η = |det|	PROVEN	TF det +1, GY det −1, lossy det = η
bits per pass = $-\log_2$|det|	PROVEN	7 of 7 cases across 3 sessions, incl. exhaustive results
n-dim residue = floor(n/2) parallel bonds	PROVEN	Exact match n = 2..12; real Schur blocks, det = λ^2
Skew half of a generator burns zero bits	PROVEN	det exp(Kt) = 1.0000000000 at all t tested
dB/dt = −tr(M)/ln2; Liouville and Landauer one statement	PROVEN	Agreement < 1e-8 at 4 times; det exp(Mt) = exp(t·tr M)
Crush is mandatory at the writer, not exceptional	FORCED	Two-layer theorem; supersedes prior OPEN status

Claim	Status	Basis
Gauge: effort-flow assignment is conventional	OPEN	No generator identified for the freedom
tr(AB) carries units of power	OPEN	Structural isomorphism only; not claimed
One ledger level = one dynamical round	OPEN	Blocks the burn/mint identity; not run
Derivation ever completed	FORBIDDEN	Complete self-description is a fixed point

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Appendix: Live Output, Quoted Verbatim

Unedited excerpts. Every load-bearing number above is reproduced here.

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STEP 1 - trace[A,B] = 8.882e-16   ||[A,B]||_F = 6.971708   product = 6.192e-15
 FAILURE. product identically 0 for every A,B, in every channel.

CHANNEL TYPE 1: THE TRANSFORMER   A -> m*A,  B -> B/m
    m       tr(AB)    ||[A,B]||   ratio ||A||/||B||   ratio/ratio_0
  1.0   4.05048962   13.709528        0.990589       1.0000
  7.5   4.05048962   13.709528       55.720621      56.2500
 20.0   4.05048962   13.709528      396.235524     400.0000
 tr(AB) invariant?  max drift = 8.88e-16
 [A,B] invariant?   max drift = 0.00e+00
 ratio scales as m^2?  56.250000 vs m^2 = 56.250000

CHAIN TEST - battery -> gyrator(motor) -> gear -> screw
 source                    24.0000    3.5000   84.00000000      6.8571
 gyrator r=0.42 (motor)     1.4700   57.1429   84.00000000      0.0257
 transformer m=7.5 (gear)  11.0250    7.6190   84.00000000      1.4470
 transformer m=0.08 (screw) 0.8820   95.2381   84.00000000      0.0093

THE TOLL   bits = log2(1/|det|)
 ideal transformer m=7.5       1.000000    0.000000    0.000000   True
 writer projection (2-to-1)    0.500000    1.000000    1.000000   True
 crush: keep low 8 of 16 bits  0.003906    8.000000    8.000000   True
 crush: keep low 4 of 16 bits  0.000244   12.000000   12.000000   True
 lossy transducer eta=0.85     0.850000    0.234465    0.234465   True

MULTIPORT   n   [A,B] skew?  floor(n/2)  distinct |eig|   match
         4     True        2           2   True
         8     True        4           4   True
        12     True        6           6   True
 bond 1: lambda=30.1512  det=909.0973  = lambda^2 ? True

CONTINUOUS   tr(M) = -0.165515   tr(S) = -0.165515   tr(K) = 0.000e+00
    t   det exp(Kt)    det exp(St)    det exp(Mt)   bits
  1.00  1.0000000000     0.84745746     0.84745746   0.23878715
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2.00  1.0000000000     0.71818414     0.71818414   0.47757429
measured bits vs -t*tr(M)/ln2 : match True at every t
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